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Review and Feasibility of Active Attitude Control and Detumbling for the UoMBSat-1 PocketQube Pico-Satellite

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Abstract—**The CubeSat form factor was very successful at reducing development and launch costs for spacecraft. The PocketQube (PQ) - one eighth the size of a CubeSat – aims to further reduce these costs. However, the question remains on whether it is feasible to implement a fully-featured satellite within the far tighter constraints imposed by a PQ.**

This paper explores the implications of including a 3-axis attitude determination and control system with provisions for desaturation and detumbling within the confines of a PQ. The hardware technology available is outlined and the applicable control laws and techniques are reviewed. This paper presents the approach that is actually being adopted in the development of the UoMBSat-1 picosatellite at the University of Malta.

I. Introduction

CubeSats currently dominate the small satellite sector and have allowed academia to participate in space engineering like never before [1]. The reasons are several [2], but primarily, it is the reduction of launch costs associated with the diminutive and standardized dimensions that has driven CubeSats forward. As miniaturization in electronic systems continues, smaller form factors like the PocketQube (PQ) [3] or the SunCube (SC) [4] should build on these advantages and gain popularity, if the many challenges can be resolved.

A 3-axis attitude determination and control system (ADCS) is currently under development for a 5×5×5cm, 250g 1p PQ at the University of Malta. The UoMBSat-1, as it is called [2], is shown as a cutout in Fig. 1 and will carry an ionospheric impedance probe (ImP) payload, currently being developed at the University of Birmingham [5].

The payload defines the 18-month mission requirements, namely: a 400-600km low earth orbit (LEO), a magnetically clean spacecraft, orbital determination and accurate attitude determination to within ±1°. Attitude control is mainly required by the communication system and GPS receiver and will be used to detumble the spacecraft soon after launch. A high rate of rotation will be established about the principal axis for passive spin-stabilization to be used as a fallback.

The axis of rotation is oriented such that the main lobe of the communications antenna (omitted in Fig 1) points at the ground station during every overhead pass. This configuration serves to distribute solar thermal flux across several faces of the satellite which results in shallow thermal cycling and reduces the reliance on any single solar panel from the five present on this PQ.

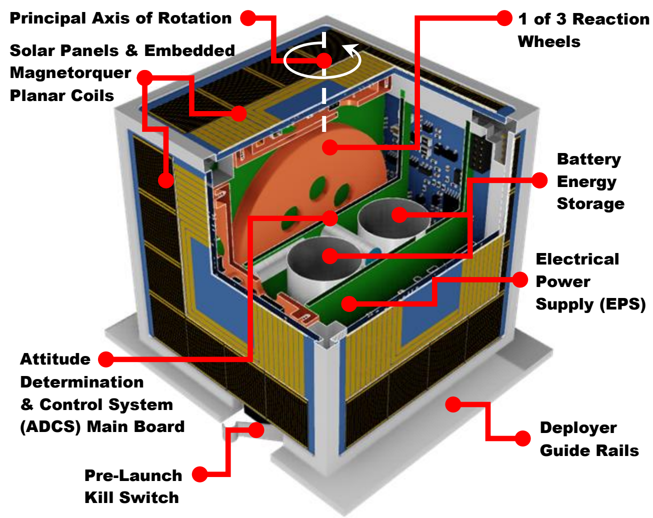

Figure 1. Cutaway drawing of the UoMBSat-1 PicoSatellite [2]

II. Attitude Disturbances on a PocketQube

External disturbances arising in the environment of the satellite will contribute to deviation from its desired attitude. For a PQ operating in LEO the disturbance torques, in descending order of importance, are of four types [6]:

1. *Magnetic Disturbance*, due to the interaction of the geomagnetic field with any residual magnetization or Lorentz forces created by electric currents in the PQ.
2. *Aerodynamic Disturbance*, due to residual atmospheric drag acting on the PQ at 7.5km/s at LEO.
3. *Gravity Gradient Disturbance*, due to uneven distribution of mass in the PQ interacting with variations in gravity across the PQ.
4. *Solar/Albedo Radiation Pressure effects*, acting unevenly on the faces of the PQ, particularly the guide rails.

Disturbances 1, 2 and 4 depend on the orbital position of the PQ, disturbance 2 depends on its orbital velocity, whilst all four depend on its orientation. Thus, for modeling and simulation purposes, the Simplified General Perturbation (SGP4) and the International Geomagnetic Reference Field (IGRF) models were used to obtain a complete characterization of the PQ disturbances. The SGP4 model [7] evaluates the orbital position and velocity of the satellite in time, whereas the IGRF model [8] is used to estimate the geomagnetic field at the orbital position provided by SGP4.

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Five reference frames are used to model the satellite dynamics and kinematics, the external torques, the position of the satellite and Earth with respect to one another, the velocity of the satellite, and the magnetic field strength of the Earth at different locations: the Earth Centered Inertial Frame (ECI), the Earth Centered Earth Fixed Frame (ECEF), the Orbital Reference Frame (ORF), the Satellite Body Reference Frame (SBRF), and the Controller Reference Frame (CRF) [9]. The external torques are defined in the ECI frame due to the frame of reference of the outputs of the SGP4 and IGRF models. The satellite dynamics and kinematics are computed in the CRF, and inputs from sensors are taken in the SBRF. Transformation between any of these frames is provided by means of quaternion algebra [10].

For actuator design purposes, worst case values were estimated using simplified models [6] and compared for a PQ sized object [11] scaled from CubeSats. A worst case residual magnetic moment of 4.4×10^{-4} Am^2 has been calculated, yielding a disturbance torque of 2.2×10^{-8} Nm, by assuming the largest contribution coming from a badly-designed solar panel current, looping round the panel's perimeter. An aerodynamic disturbance of 1.6×10^{-8} Nm was estimated from the residual atmosphere at 550km acting on a total incident area of 40cm^2 at 7586m/s, at a moment arm taken at ⅓ the size of the 70cm band antennas. Gravity gradient and solar pressure effects were deemed negligible at 9.0×10^{-10} Nm and 4.1×10^{-10} Nm respectively.

III. Power, Mass and Volumetric Budget

The entire PQ must work within an average orbital power budget of about 300mW. This figure takes into account the efficiencies of the solar panels, the distributed maximum power point trackers (DMPPT), the electrical power supply (EPS) and a 25% margin for degradation. After subtracting the power requirements of the more critical subsystems, a 50mW power budget was dedicated to the ADCS. In the same manner, no more than 100g of mass budget can be dedicated to the ADCS. Any ADCS sensing and control must all fit on the centre-most 40x40 mm PCB in the system, while any actuators may be distributed for optimal positioning if they do not interfere with other systems.

IV. Suitable Actuators

Due to the magnetic cleanliness requirement, passive control techniques such as permanent magnets and hysteresis rods must be avoided. Gravity booms and solar sails are too mechanically complex to fit in a 1p PQ, so active ADCS approaches are necessary. The following were considered:

1. 6PPT: 3 Micro Pulsed Plasma Thruster (µPPT) pairs [12]
2. 3MT: 3-axis copper coil magnetorquers (MT) [13]
3. 3RW: 3-axis reaction wheels (RW) [14]
4. 3RW3MT: 3-axis RW with 3 axis MTs [14]

µPPTs have been successfully used in CubeSats for the generation of thrust as well as torque [12]. However, the need for compact high voltage transformers is somewhat in conflict with the magnetic cleanliness requirement. The technology has yet to be successfully demonstrated at the PQ scale. µPPTs were included in the WREN PQ [15] in 2013. However, sustained contact with this PQ had not been established and no data is available. Electromagnetic Interference (EMI) is also likely to be a major problem and much more development is required before this can become a viable option for precision attitude control for PQs.

A 3MT solution is the most magnetically clean, given that coreless MT coils can be energized when the ImP is inactive. However, such actuation would be rank deficient since the coil whose axis is parallel to the earth's magnetic field cannot produce any torque. Moreover, any coil close to parallel would need substantial overdrive to generate any useful torque. This means that reliance on just MTs requires them to be sized and rated to provide at least 10 times the dominant disturbance torque, and this may have to be increased to a 100-fold for reasonable agility while factoring-in all the uncertainties. Coupled with the necessary absence of a magnetic core, these requirements result in large and heavy coils that consume excessive power.

An alternative method of actuation is to use the power-efficient 3RW setup. However, in practice this suffers from speed saturation of the RWs. The 3RW3MT configuration addresses the latter problem, but with much smaller MT coils than 3MT alone. This is because the MTs can now be opportunistically activated only when the B-field is at ~90º to the coil axis for RW desaturation only. This lends itself to much greater overall power and mass efficiency, while the RWs can handle the bulk of the actuation with more accurate instantaneous control (see Table 1).

On the other hand, the volumetric constraints restrict motors to brushed DC coreless variants which typically carry a tiny annular NdFeB permanent magnet which, although screened with a mild-steel canister, does leak some magnetic flux. The measured flux density at the motor surfaces is quite low (~ 200µT) and drops to 10% and 1% of the geomagnetic flux density (~50µT) within 2cm and 5cm respectively, but it has yet to be seen to what extent this will distort the field in proximity to the ImP antenna. That said, removal of all magnetic disturbances will be impossible because even the current in the solar panels and the batteries will create some disturbance. Therefore, some method of compensation for the payload's measurements would have to be devised anyway.

Maximizing the RW moment of inertia is required to improve actuation torque at low rotor acceleration and hence reduce brush wear. An optimal flywheel configuration therefore requires peripheric concentration of mass with a lightweight support structure that can withstand the peak centrifugal forces. This can be readily achieved with a bimetallic structure such as the one shown in Fig 2. In this case, dense 0.5mm tungsten wire was inserted into a circumferential slot of high aspect ratio in the rim of a Ø37mm aluminum flywheel, leading to a moment of inertia of 1.2 kg.mm^2 with a mass of only 5.5g. This brings the total 3RW actuator mass, including the motors, to under 20g.

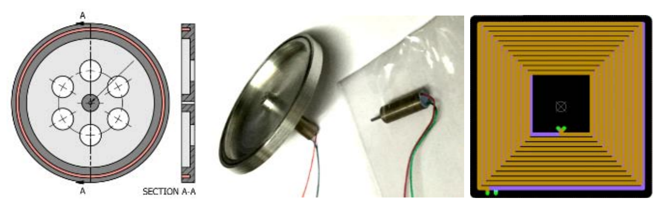

Figure 2. Bimetallic reaction wheel, core-less motor and MT planar coil

TABLE I. COMPARISON BETWEEN OPTIMISED DESIGNS OF THE TWO ADCS ACTUATION TECHNIQUES

Quantity	3MT	3RW3MT	Units	Notes
Peak Disturbance Torque (τ)	3.7×10^{-8}		Nm	contribution of magnetic, aerodynamic, gravity gradient, solar pressure effects
Target Actuation Torque (T)	3.7×10^{-6}		Nm	target actuation T = 100τ (MT actuation calculated at 90º to B-field of 50μT)
PQ Moment of Inertia	104		kg.mm^2	assuming 50mm cube of 250g with uniform distribution of mass
PQ Angular Acceleration	73		rpm/min	achieved with Target actuation T = 100τ, when PQ is spinning at 600 rpm
MT Conductor cross section	0.051	0.0035	mm^2	taking into account 3.3V supply rail voltages
MT Winding Size (L×W)	40×40	31×31	mm	taken at the mean current centre line
MT Winding Turns /Axis	361	352 × 2	#	taking into account 3.3V supply rail limitations
MT Winding Mass /Axis	26.5	1.14 × 2	g	ignoring FR4 Mass which is needed anyway for solar panel rigidity
MT Winding Resistance	24.9	227.8	Ω	assumed to be operating at 100 ºC
RW Flywheel Moment of Inertia	-	1.22	kg.mm^2	W/Al - bimetallic, radius = 18.5mm, rim = 2.5mm , rim thickness = 3mm
RW Flywheel Acceleration	-	1720	rpm/min	for actuation = peak disturbance torque × 99
RW Motor + Flywheel Mass /Axis	-	6.17	g	tungsten-aluminum bimetallic composite flywheel
Total ADCS Actuator Mass	79.4	25.3	g	including all three axes, reserving a 20g budget for the ADCS electronics
Actuator Energy Efficiency	0.06	5.4	%	to generate target actuation torque, (PQ spinning at 600 rpm, RWs at 2000 rpm)
Expected Actuator Life Span	∞	>10000	hours	extrapolated from accelerated motor wear test data
Maximum ADCS Actuator Power	1234	43 + 287	mW	(RW+MT) delivering a combined 100τ of torque per axis, RWs at 2000 rpm
Average ADCS Actuator Power	7.9	20 + 5.8	mW	(RW+MT) countering peak disturbance torque in one axis, RWs idle at 2000 rpm

V. ADCS HARDWARE ARCHITECTURE

In the UoMBSat-1, two of the RW flywheels are mounted parallel to the solar panel faces at very close tolerances as shown in Fig 3. This limits any bending (due to launch vibration) to within the elastic limit of the Ø700μm motor shafts. Space utilization is maximized by ensuring that the DMPPT components fit within the flywheel cavities. The third flywheel is mounted parallel to the ADCS board onto which all three motors are attached. Heat is transported away from the motors via conduction through the ADCS board. (Fig. 4) This is an important consideration in vacuum and helps limit large changes in armature resistance and the motor model.

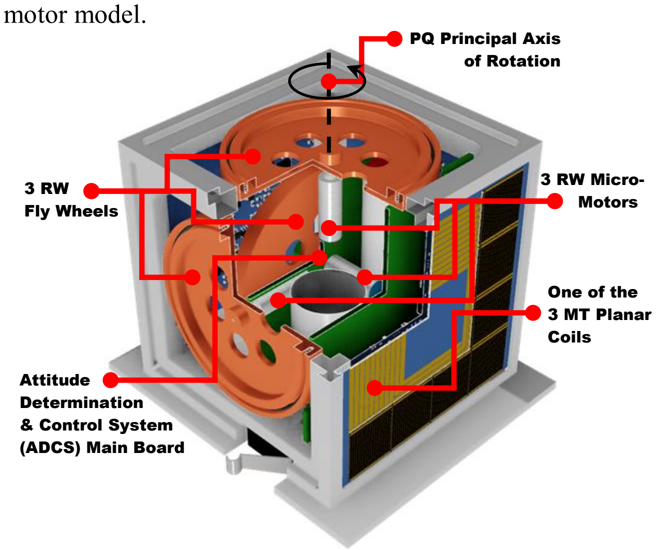

Figure 3. Cutaway drawing showing reaction wheel configuration

The ADCS will draw its power from two lithium-ion-polymer (LiPo) cells connected in parallel which can provide the high instantaneous currents required for the MTs and other PQ functions. This choice of energy storage was based on the fact that LiPo cells have a higher energy and power density compared to other technologies. Soft case cells were chosen to avoid the steel cases for magnetic cleanliness. In order to maximize the mission lifetime, the cells had to be chosen such that they will have minimal degradation with the given operating conditions. Preliminary tests have been carried out on a number of different cell types. These tests consisted of monitoring the cell capacity in ambient conditions, vacuum conditions and at low temperatures (-15°C) by performing a number of charge discharge cycles. Cylindrical cells did not show any loss in capacity under vacuum. In fact, no layer delamination was observed. The cylindrical cells tested also showed minimal capacity loss at low temperatures. Cell positioning was selected such that the cells experience reduced temperature excursions while maximizing the inertia about the principal axis of the PQ. The next stage will be to test the chosen cells in long term charge-discharge cycles, to observe the capacity loss pattern.

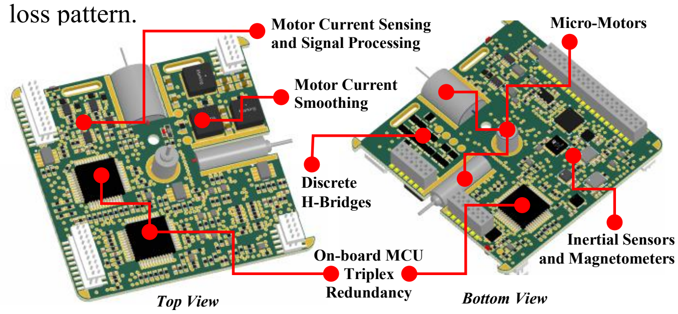

Figure 4. ADCS main board with all three micro-motors

The ADCS requires tri-axial feedback data from an inertial measurement unit (IMU) and a magnetometer in order to determine the PQ's attitude. The most viable solution to implement these sensory units is to utilize microelectromechanical systems (MEMS). Several CubeSat missions [16]-[19] have demonstrated the adequacy of commercially off the shelf devices (COTS) for LEO missions. These units typically incorporate several inertial measurement instruments (accelerometer, gyroscope, magnetometer…) in a single chip-scale surface mount package. Recent fabrication improvements have optimized the power consumption, accuracy and precision of such devices to supply the ever-growing smartphone, wearables and IoT (Internet of Things) industry. This trend is also observable for microcontrollers (MCUs) and other integrated circuits (ICs) and the diversified and abundant market has made the implementation of an active, fully-controllable ADCS with dependable measurements evermore feasible.

The COTS devices in question however, do not guarantee reliable operation at elevated levels of ionizing radiation such as those encountered in LEO. ICs are prone to experience single event effects (SEEs), with the most common and disruptive being single event latch-ups (SELs) and single event upsets (SEUs). SELs describe the switching on of parasitic thyristors in ICs when a high energy electron traverses a PN-junction liberating an electron-hole pair. The thyristors tend to short the supply rails hence hindering the operation of other devices as well. SELs can potentially cause permanent damage and IC failure if not handled. SEUs, also referred to as soft errors, have a similar mechanism to SELs however result in a state change in CMOS devices [20][21]. MCUs are the most susceptible devices to encounter SEUs due to their high level of integration and high component count, and further device scaling efforts will only increase their susceptibility to SEEs [22][23]. This can potentially jeopardize the functionality of the ADCS and the need for a redundant design is therefore apparent. The hardware must provide enough sensory inputs and recovery options for the software to monitor and choose from, such that the failure in one device does not critically hinder the functionality of the whole system.

The ADCS hardware should incorporate redundant sensory units to be able to cross check the health of each unit. These units should be distributed along the board and accessed through separate interfaces, which ensures that the design does not succumb to a single point of failure. Such distribution of resources should also be applied to the processing unit, hence a multi-processor democratic approach is preferred over a centralized master-slave one. Finally, the choice of redundant components should span multiple manufacturers and different technologies, such that the probability of common-mode failure is reduced. This diversified approach ensures that the susceptibility of one COTS device to extreme environmental conditions do not propagate throughout the system.

The ADCS board also provides the power electronics for driving the various actuators. The 3RW3MT configuration opted for in this project, requires the ADCS board to also provide a mount for the motors to couple to the pico-satellite's chassis. A switching drive (H-bridge) for the motors and MTs is considered the most efficient driving methodology. On the other hand, switching frequencies and snubbing networks must be chosen carefully so as not to excite the self-resonant modes of the MT coils. Snubbing circuits needs to be inserted judiciously to avoid the MTs from radiating too much EMI.

The H-bridge circuitry to drive the motors can be implemented on board the ADCS. However, since the MTs are located at the extremities of the PQ, the driving circuitry will be housed onto the peripheral panels rather than on the ADCS board. This design consideration limits EMI generated by the interconnecting media. To further prolong the functionality of the ADCS in orbit, the driving signals of the actuators should also be provided from different sources.

A modular, scalable architecture is being proposed to satisfy the criteria presented. Fig. 5 shows a block diagram representation of this architecture, implementing health monitoring and redundancy on one MCU. The design can easily be extrapolated to monitor all MCUs. Triplex redundancy can be realistically accommodated on the ADCS board and allows for democratic MCU health scrutiny.

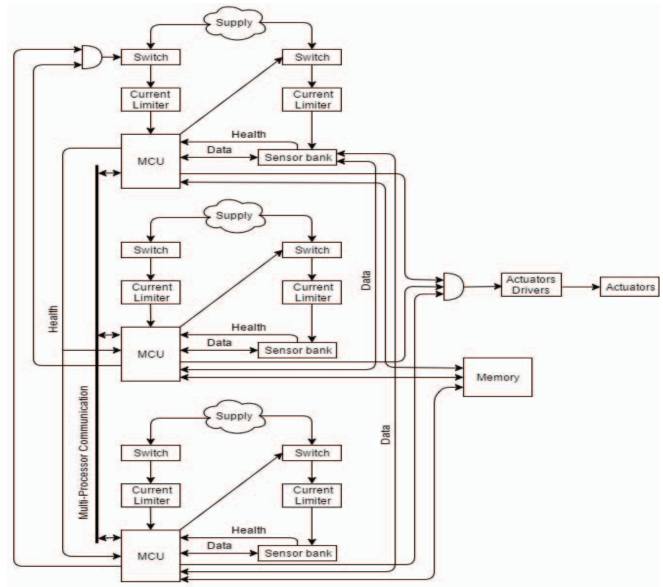

Figure 5. Proposed ADCS hardware block diagram.

The following points highlight the functionality presented in the architecture of Fig. 5:

- Current limiters are required to detect and clear SELs. They should be constructed from discrete devices rather than ICs such that they do not suffer SELs themselves.
- MCUs are responsible to monitor the health of their sensor banks. This can be done by measuring the supply rail currents of the sensors.
- The MCUs have access to the other sensor banks with the permission of the other MCUs. This allows sensors to keep on operating even if their associated MCU fails.
- An inter-connecting bus allows the MCUs to share data and facilitates parallel processing.
- An unresponsive MCU can be ruled out democratically if remaining MCUs agree that it is under fault condition.
- Redundant MCUs can confirm that the driving MCU is producing the correct actuation, if not the driving signal is gated and all faults are logged in external memory.

VI. ATTITUDE CONTROL STRATEGY

As outlined previously, control of satellite attitude is achieved using two different types of actuation; three axis-oriented motor-driven RWs, and three axis-oriented MTs. The use of RWs allows for precise, yet quick and energy efficient changes in attitude, whilst the MTs are used for desaturation of the RWs, detumbling, and possibly, slow changes in attitude in case of motor failure.

The controller design methodology to be adopted is similar to [9] , where the dynamic and kinematic models of the satellite as well as the external torques and satellite position models are formulated and simulated on Simulink [24]. Secondly, design of the controller is performed and simulated on the satellite model. Finally, the controller, sensors and actuators will be implemented in hardware, then fully characterised and tested.

The selection of the controller is based on three requirements which are accuracy, robustness and optimized energy consumption. Multiple controllers will be analyzed and the most suitable controller (or a combination thereof) will be chosen according to the given requirements. The classical strategy to reduce power consumption, yet maintaining accurate control, is an optimal control approach such as the linear-quadratic regulator (LQR), as in [9] and [25], or the linear quadratic Gaussian (LQG) regulator which compensates for measurement and process noise. Model Predictive Control could also be a viable alternative.

For robustness purposes under model imprecision, the H-infinity control approach would be a strong candidate. The advantage in using such a controller lies in the fact that stable attitude control can be achieved without the need for a precise model of both the satellite dynamics and the external disturbances. In order to compensate for a possibly reduced control accuracy, an H-infinity Loop Shaping approach, which gives robust control and accurate performance, is often preferred [26] .

Another robust methodology to be considered is Sliding Mode Control. The advantage of this approach is that by applying a non-continuous high frequency signal to the actuators, a high order, nonlinear dynamic system will behave like a linear, reduced order dynamic model that is independent from the system's dynamic model, making the sliding mode controller robust against external disturbances and model inaccuracies [27]. However, the use of high frequency discontinuous actuation limits the use of this controller to the magnetorquers only and hence, this control would find good use in case of reaction wheel/motor failures. A sliding mode controller using MTs only is described in [28] where a three dimensional manifold was proposed and shown that the satellite motion on the manifold is asymptotically stable

VII. Desaturation Strategy

Due to external disturbance torques, the RW's angular velocity tends to increase and possibly reach saturation. This would limit the attitude control action and increase motor wear and tear. Hence, a desaturation strategy is needed. A commonly used method for wheel desaturation is the cross product desaturation control law (a momentum unloading controller) [29]. This controller is implemented simply by making the magnetic dipole moment of the magnetorquers proportional to the cross product between the reaction wheel's angular velocity and the measured earth's geomagnetic field.

A possibly more efficient way for wheel desaturation was introduced in [30], where combined reaction wheel control and desaturation was achieved by the use of a periodic, time-varying LQR controller. Following the same methodology, it should also be possible to realise an LQG controller where, by the careful selection of the weighing values in the cost function R matrix, both the RW angular velocity and the MT current can be reduced to acceptable values, hence keeping energy consumption and motor wear and tear to a minimum.

VIII. Detumbling Strategy

As the PQ is released in LEO, it usually starts to tumble. Hence, a detumbling controller is needed. The simplest and most commonly used strategy for detumbling control is the B-dot controller [29]. This controller sets the value for the magnetic dipole to be generated by the MTs to be proportional to the rate of change of the geomagnetic field (hence the term B–dot). A variation of this controller is the Bang-Bang B-dot controller where in [22] it is analytically shown that it achieves around 27% higher torque at the expense of decreased energy efficiency. All these controllers suffer from the fact that satellite detumbling can never be achieved entirely.

An alternative method was proposed in [31] based on satellite angular velocity feedback, which showed that in the presence of a time varying magnetic field, global asymptotic stability is achieved from an arbitrary tumbling condition to zero angular velocity. In [32], all the above mentioned detumbling controllers were simulated and it was shown that the controller proposed in [31] obtained a faster exponential decay, with the satellite angular velocity converging to zero.

IX. Preliminary Results for De-tumbling Control

For the purpose of simulating the detumbling controller, a satellite model was implemented in Simulink [24]. This models the satellite in low earth orbit. The orbit parameters were obtained using the SGP4 model [7]. The disturbances experienced by the satellite were modeled as being due to the aerodynamic drag, gravity gradient, solar radiation and the earth's geomagnetic field obtained using the IGRF model [8].

The B-Dot and the Bang-Bang B-Dot controllers described in Section VIII were simulated as continuous-time controllers. Results are shown in Fig.6 where the solid lines correspond with the B-Dot controller and the dashed lines with the Bang-Bang controller, both assuming an initial satellite angular velocity of [0.1, -0.005, -0.01] rad/s. The simulations show that both controllers can obtain comparable results and are able to stabilize the satellite's angular velocity norm to less than 0.006 rad/s in roughly 5400 sec which corresponds to one orbit. The choice on which controller to use is then to be based on power and circuit complexity constraints where, although the Bang-Bang controller drive circuitry is easier to design, its magnetic moment is fixed and cannot be controlled.

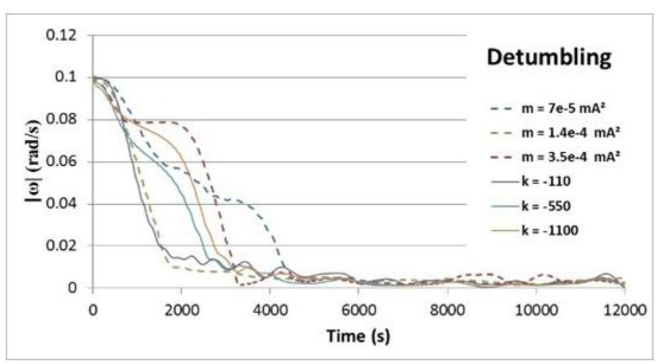

Figure 6 – B-Dot and Bang-Bang B-Dot Simulation Result

X. PRELIMINARY RESULTS FOR ATTITUDE CONTROL

In order to characterize the performance of the attitude controller as a proof of concept, it is planned to implement a physical prototype of the satellite operating on Earth. This will be supported by a rigid attachment and the "satellite" is allowed to rotate about the vertical axis only so as to cancel the effects of gravity as if it were in space. The dynamics of this set-up were derived and appropriate controllers were designed and simulated.

The controller opted for this system is a linear-quadratic tracker (LQT) i.e. an optimal linear quadratic controller with integral action [33]. For this system, cascade control is implemented, where the outer loop controlling the desired attitude demands a torque, and a faster inner loop controls the motor torque accordingly to provide the demanded torque. For the inner loop, a proportional-integral (PI) controller was used. Both outer and inner loop controllers are operating in discrete-time at a sampling rate of 40 Hz and 8kHz respectively.

Fig. 7 shows the response of the control system having zero initial conditions, while requesting a desired angle of 5 degrees. This desired angle is reached in approximately 10 seconds. LQT, permits the designer to find a compromise between the demanded torque input from the motors and the desired angle of the satellite according to the selection of the weighting matrices in the cost function of the controller. Thus a faster response could be obtained at the cost of an increased demanded torque, and hence energy consumption.

The ringing during the first few seconds of the torque and voltage responses of 7 is due to the phase lag introduced by the 4th order Bessel filters used for anti-aliasing which were also modeled in the simulation.

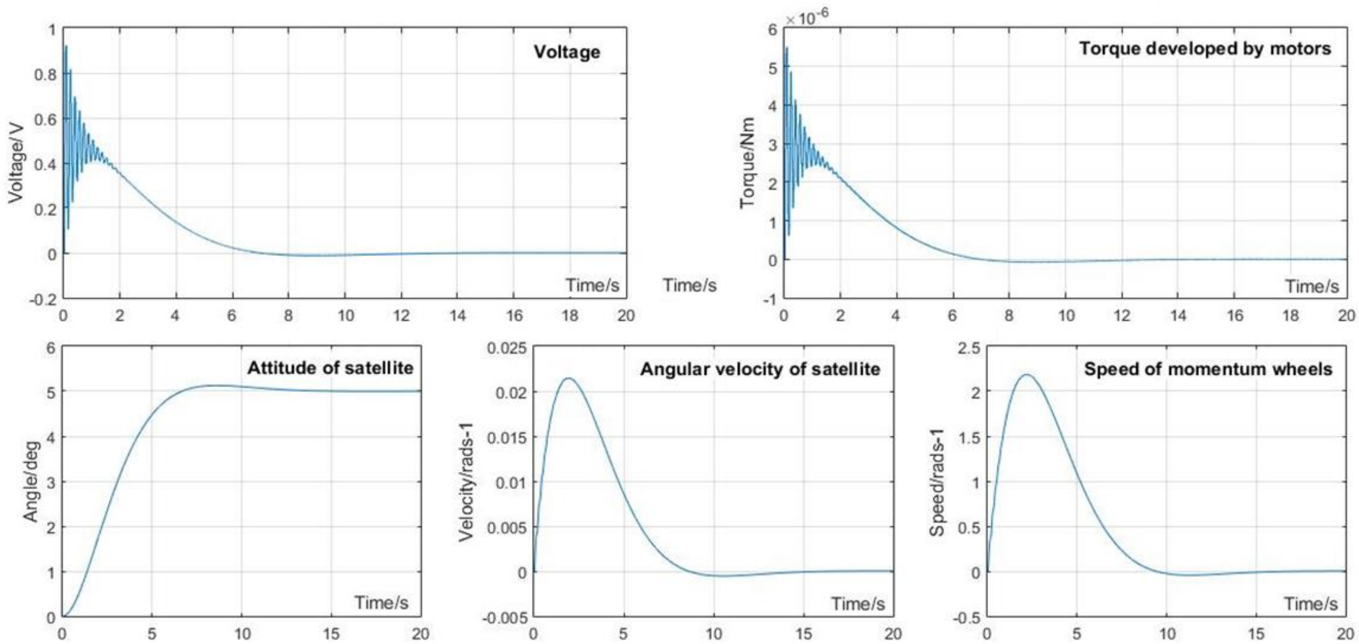

Figure 7 - Response of the LQT controller for the satellite prototype model

XI. FUTURE PLANS

Preliminary reliability testing has been conducted on several components and the results are encouraging, but this currently lacks statistical significance. Much more will have to be done in this domain to allow the judicious selection of operating modes that will extend the useful life of the PQ.

The actuators, particularly the motors, will eventually wear out and fail. However, this need not spell the end of the mission and hence presents several interesting avenues for further research. A number of fallback operating modes are being considered given that the 3MT3RW is essentially over actuated. The MTs can still be used for attitude control albeit inefficiently.

One of the requirements of the ADCS system is to achieve an accuracy of 1° and hence, a reliable and accurate attitude determination strategy is also needed. Given the size constraints of the PQ, sensor usage is physically limited and the most viable solution would be to use sensors which are already present on the PQ satellite due to their need for the mission. This motivates the use of the solar panels and magnetometers as sensors and fuse their measurement to obtain a reliable attitude estimate.

One of the most researched methods to obtain a reliable estimate of a satellite's attitude is the Extended Kalman Filter (EKF) [35]. The EKF is essentially a Kalman Filter (KF) [34] applied to non-linear systems by making use of system linearization. The EKF is vastly used for sensor data fusion and in [35] , an EKF is used to fuse 3-axis magnetometer data and 2-axis solar panel data to obtain a reliable attitude estimate where it is shown that by having a low sun measurement noise, an accuracy of 2.5° is achieved. Although this is a very good result, it is still not within the mission's requirement. Reasons for these inaccuracies are related to the linearization method used in the EKF where it only achieves first order accuracy.

Given the 1° accuracy requirement, an improved version of the EKF is needed. A candidate for this is another extension of the KF called the Unscented Kalman Filter (UKF) [36], [37]. The improvement in this filter lies in the method of linearization. As opposed to the EKF, where the state distribution is approximated by a Gaussian random variable and then propagated using the first order linearization of the non linear system, the UKF uses the principle that a set of discrete points can be used to get the mean and variance. In [37], it is shown that the UKF yields a performance which is equivalent to that of a KF for linear systems and superior to that of an EKF with the advantage of being much less difficult to implement than an EKF. In [38], the UKF is used for spacecraft attitude determination and is shown to achieve performance which exceeds that of a standard EKF for large initialization errors.

XII. Conclusions

The rapid degree of miniaturisation that is being brought about by the massive adoption of portable and wearable devices is clearly working to the advantage of small satellites. The cost of the components is very reasonable and reliability can still be achieved with careful design. The energy requirements of modern MEMS sensors and fast microprocessors also fall well within the power budget of the solar panels that can be accommodated on a PQ.

In conclusion, a fully featured ADCS for use in PQs appears to be technically feasible using commonly available COTS components and materials, and one of the purposes of UoMBSat-1 is to demonstrate this in a real mission.

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